

ECHOFLOW AI

Physics-informed acoustic flow-state monitoring for manual peritoneal dialysis

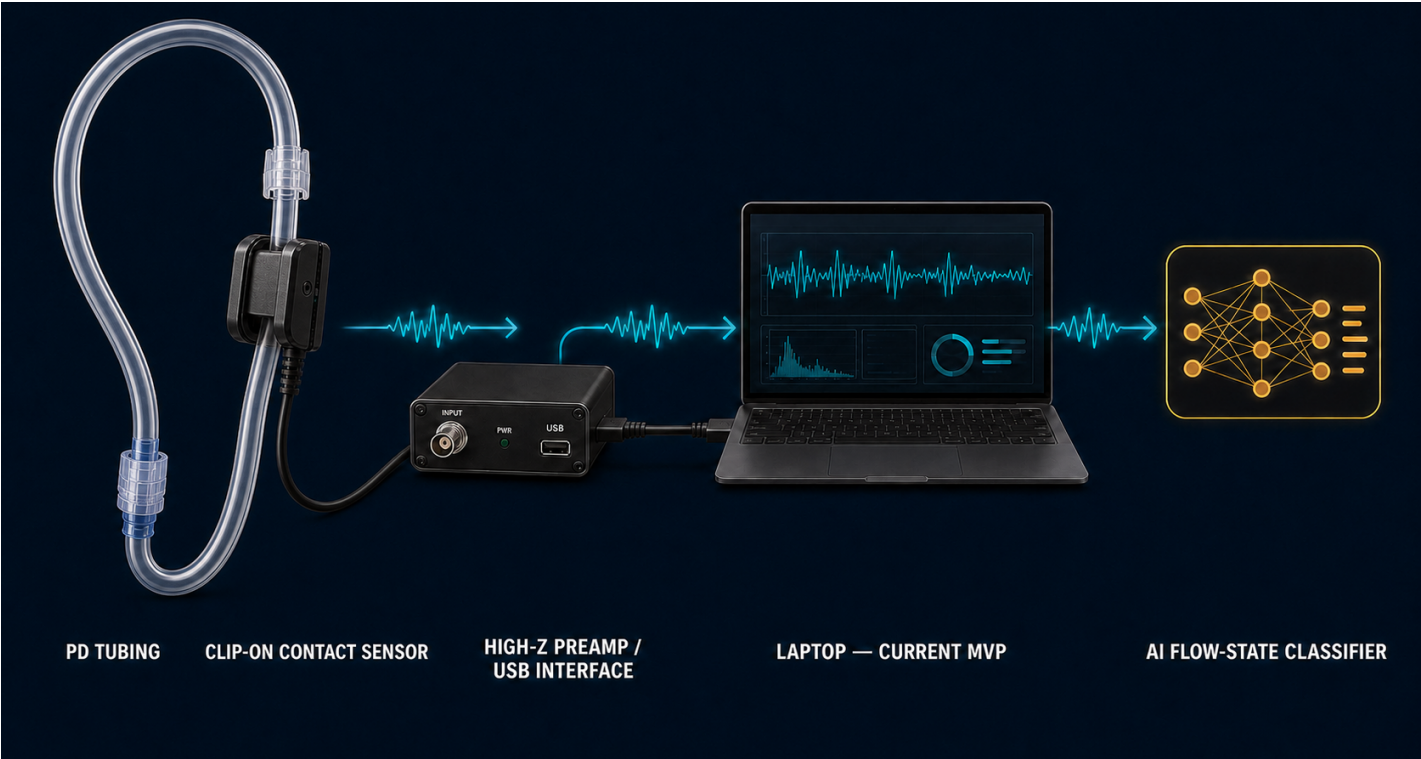
Digital proof of concept | MATLAB/Simscape + Python | Joel Choy, Imperial College London

RESEARCH HYPOTHESIS Controlled flow restriction changes tube-wall vibration signatures enough to support external flow-state classification.

01 CURRENT DIGITAL PROTOTYPE

1 SIMSCAPE hydraulic traces	2 PYTHON acoustic proxies	3 DSP 233 features	4 RANDOM FOREST state probabilities
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Hydraulic model. Simscape generated normal, restricted, severely restricted and stopped flow traces. Acoustic proxies. Python converted them into labelled 16 kHz, 3 s synthetic audio windows. Targets. Normal, reduced flow, stopped flow and a non-flow noise control. A held-out synthetic restriction example was reported as reduced flow with 84% confidence.



Proposed physical architecture. Current MVP: laptop-based simulation, signal processing and classification. Sensor validation is the next evidence gate.

02 SIGNAL PROCESSING AND MACHINE LEARNING

Feature vector. 233 descriptors per window: amplitude and RMS summaries; crest factor and kurtosis; zero-crossing rate; spectral shape; 20 MFCCs plus deltas; Welch PSD, spectral entropy and band-power ratios; Hilbert-envelope statistics. Model. Random Forest captures nonlinear interactions between tabular features, requires little scaling and provides feature importances. Session-level holdout prevents windows from the same synthetic session appearing in both training and testing.

03 EVIDENCE BOUNDARY AND NEXT TEST

DEMONSTRATED Digital pipeline. Hydraulic traces to synthetic acoustics, feature extraction, classification and probability output. Research output only. No claim of obstruction diagnosis, dialysis adequacy or treatment recommendation.	NEXT EVIDENCE GATE Controlled tubing trial. Fixed contact sensor plus high-impedance preamp; repeated normal, restricted and stopped trials with remounting and position checks. Success criterion. Within-state recordings remain similar while held-out states remain distinguishable.
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04 ENGINEERING CHOICES

PHYSICS-INFORMED FIRST Tests the data and inference pipeline before any claim about sensor performance.	INTERPRETABLE BASELINE Random Forest suits nonlinear tabular features and exposes feature importances.	EXTERNAL SENSING The proposed reusable clip remains outside the fluid pathway.
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